



Level 2

Challenge B



Directions:

In the `prolog_challenges.pl` file, create a predicate called `"list_rindex_sum"` that does the following:

The predicate will take in 3 arguments: `List` - a list of lists of numbers, `RIndex` - a 0 based index starting at the right (end of the list), `Sum` - the sum of the sublist at the `RIndex`. You will need to create your own `list_sum` predicate to help with this. To help understand this predicate, consider the following potential values for the parameters. If `List` is the list `[[1,2,3,4], [5,6,7], [8,9], [10]]`, then the following holds:

List	[	[1,2,3,4],	[5,6,7],	[8,9],	[10]	]
RIndex		3	2	1	0	
Sum		1+2+3+4 = 10	5+6+7 = 18	8+9 = 17	10	

Examples:

```
/* query */  ?- list_rindex_sum([[1,2], [3,4], [5,6]], 0, Sum).
/* result */ Sum = 11 .
/* note: RIndex is 0 and so we look at the sublist at index zero from the
right (i.e. last sublist). The sum is 5+6 which is 11. */

/* query */  ?- list_rindex_sum([[1,2], [3,4], [5,6]], 1, Sum).
/* result */ Sum = 7.

/* query */  ?- list_rindex_sum([[1,2], [3,4], [5,6]], 2, Sum).
/* result */ Sum = 3.

/* query */  ?- list_rindex_sum([[1,2], [3,4], [5,6]], 3, Sum).
/* result */ false.

/* query */  ?- list_rindex_sum([[1,2], [3,4], [5,6]], -1, Sum).
/* result */ false.

/* query */  ?- list_rindex_sum([[10], [18,9,100]], 0, Sum).
/* result */ Sum = 127.

/* query */  ?- list_rindex_sum([[10], [18,9,100]], 1, Sum).
/* result */ Sum = 10.

/* query */  ?- list_rindex_sum([], 0, Sum).
/* result */ Sum = 0.

/* query */  ?- list_rindex_sum([], 0, Sum).
/* result */ false.

/* query */  ?- list_rindex_sum([[1,2,4], [5,3,2]], 1, 7).
/* result */ true.
```